A Project Report

On

IDENTIFICATION OF COUNTERFEIT PRODUCTS USING BLOCKCHAIN TECHNOLOGY

Submitted in Partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

SUBMITTED

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2024-2025

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CERTIFICATE

This is to certify that the project report entitled **"IDENTIFICATION OF COUNTERFEIT PRODUCTS USING BLOCKCHAIN TECHNOLOGY"** being submitted to the Department of Information Technology, J.B. Institute of Engineering and Technology, in accordance with Jawaharlal Nehru Technological University regulations as partial fulfilment required for successful completion of Bachelor of Technology is a record of bonafide work carried out during the academic year 2024-2025 by,

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DECLARATION

We hereby certify that the Major project report entitled "**Identification of Counterfeit Products Using Blockchain**" carried out under the guidance of **G. Kumar**, **Assistant Professor** is submitted in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Information Technology**. This is a record of bonafide work carried out by us and the results embodied in this major project report have not been reproduced or copied from any source. The results embodied in this project have not been submitted to any other University or Institute for the award of any other degree or diploma.

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ACKNOWLEDGMENT

At outset we express our gratitude to almighty lord for showering his grace and blessings upon us to complete this Major Project. Although our name appears on the cover of this book, many people had contributed in some from or the other to this project development. We could not have done this Project without the assistance or support of each of the following.

First of all we are highly indebted to **DR. P. C. KRISHNAMACHARY,** Principal for giving us the permission to carry out this Major Project.

We would like to thank **DR. A. RAMESH BABU**, Associate Professor & Head of the Department of INFORMATION TECHNOLOGY, for being moral support throughout the period of the study in the Department.

We are grateful to **G. Kumar**, Assistant Professor of the Department of INFORMATION TECHNOLOGY, for his valuable suggestions and guidance given by him during the execution of this Project work.

We would like to thank Teaching and Non-Teaching Staff of Department of Information Technology for sharing their knowledge with us.

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ABSTRACT

In this project, Blockchain technology is employed to authenticate products, addressing the issue of counterfeit items in the supply chain. Counterfeit goods, such as fake medicines, can lead to significant financial losses and pose a threat to public health. The challenge arises when third-party distributors clone barcodes or manufacture counterfeit products, potentially undermining the trust in online transactions. Blockchain offers a solution by eliminating the need for third parties and providing a tamper-proof verification system. Product details, including barcodes, are converted into digital signatures and stored on the Blockchain, ensuring their integrity. Any unauthorized alteration of data is immediately detected due to the decentralized nature of Blockchain, where each transaction is verified across multiple servers. The system's design includes modules for both admin and user interactions, allowing for the addition, viewing, and authentication of product data.

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CHAPTER 1

INTRODUCTION

1.1 Aim:

The aim of this project is to develop a secure and efficient system for authenticating products and preventing counterfeit products in supply chains using Blockchain technology. By leveraging the tamper-proof and decentralized nature of Blockchain, the system aims to ensure product authenticity and prevent fraudulent transactions without relying on third-party intermediaries.

1.2 Objectives:

1. **Prevent Counterfeit Products**: Identify and eliminate fake products, such as counterfeit medicines, through Blockchain-based authentication.
2. **Enhance Data Integrity**: Ensure the immutability and integrity of product data stored in the Blockchain.
3. **Decentralized Verification**: Eliminate the need for third-party intermediaries for transaction verification by using Blockchain algorithms.
4. **Increase Trust**: Build trust among stakeholders in the supply chain by providing a reliable mechanism for product authentication.
5. **Real-Time Detection**: Enable instant detection of counterfeit barcodes or altered product details.

1.3 Scope of Study:

1. **Supply Chain Management**:
    - Focuses on preventing the proliferation of counterfeit products in the supply chain by authenticating products at various distribution levels.
2. **Healthcare and Pharmaceuticals**:
    - Prevents fake medicines from entering the market by verifying their authenticity, thereby reducing potential risks to human lives.

3. **E-Commerce and Online Transactions**:
    - Reduces reliance on third-party entities for transaction verification and prevents fraud in online payments.
4. **Blockchain Implementation**:
    - Demonstrates the use of Blockchain technology, specifically Ethereum, for storing, retrieving, and validating product data using Solidity-based smart contracts.
5. **Fraud Detection and Prevention**:
    - Highlights Blockchain's ability to detect altered data and prevent future malicious activities through distributed data storage and hash code verification.
6. **User Interaction**:
    - Provides modules for both administrators and users to interact with the system, ensuring secure storage and retrieval of product data.

1.4 Introduction:

Counterfeit products pose a significant challenge across industries, particularly in supply chains where products pass through multiple third-party distributors. These counterfeit products, especially in critical areas such as healthcare, can lead to financial losses and jeopardize lives. For instance, fake medicines manufactured with counterfeit labels can have devastating consequences. Traditional systems rely on intermediaries to authenticate products and facilitate transactions, but these intermediaries often become points of vulnerability, leading to fraud or data misuse.

Blockchain technology offers a revolutionary solution to these challenges by providing a decentralized, secure, and tamper-proof method of storing and verifying data. Unlike traditional systems, Blockchain does not depend on third-party trust; instead, it relies on cryptographic algorithms and distributed ledgers to ensure data authenticity and integrity.

In this project, Blockchain is used to convert product details and barcodes into digital signatures, which are stored on a Blockchain server. This ensures that any alteration to the data is instantly detectable. For example, Blockchain's distributed nature ensures that the same transaction data is stored across multiple servers. If a

malicious user alters data on one server, the hash code changes, and the discrepancy is detected by comparing it with data from other servers. This makes Blockchain highly resistant to tampering.

The global development of the product always comes with risk factors such as counterfeiting and duplication which in turn can affect the company name, reputation, revenue and customer satisfaction. The trading and marketing of counterfeit products is growing in humongous. It affects the sales, reputation, and profits of the companies and also do poses a fatal threat for the unsuspecting buyers. In order to ensure the identification and traceability of false goods or products throughout the supply chain and to combat this phenomenon, a fully functional blockchain system is proposed.

Each block in the chain contains multiple transactions, and every time a new transaction occurs on the blockchain, a record of that transaction is added to every participant's record. The decentralized database managed by the number of participants is known as Distributed Ledger Technology (DLT). Blockchain is a type of DLT in which transactions are recorded with an immutable cryptographic signature called a hash. Blockchain technology helps to solve the problem of counterfeiting a product. Blockchain technology is more secure compared to other technology.

Once the product is stored on the network hash code is generated for that product and it is possible to maintain all transaction records of the product and its current owner as a chain will be created for that product transactions. It will store all the transaction records as blocks in the blockchain. In the proposed system we are assigning a generated QR code or barcode to a particular product created by manufacturer along with all the details of the product and the end customer can scan that QR code to get all information about that product. After scanning the QR code or barcode the user can identify whether the product is real or fake

The proposed system comprises multiple modules for managing and authenticating product details. Administrators can add product details, generate digital signatures from barcodes, and store this information on the Blockchain. Users can retrieve product data and authenticate barcodes by comparing the stored digital signatures. The system uses Ethereum Blockchain and Solidity programming for smart contracts, ensuring secure and efficient handling of product data.

1.4.1. Basics of QR Code:

A **QR (Quick Response) code** is a type of matrix barcode (or 2D barcode) that stores information in both horizontal and vertical directions, enabling it to hold more data compared to traditional barcodes. QR codes are widely used for various purposes, including product tracking, authentication, and marketing. A QR code consists of a series of black squares arranged in a square grid on a white background. These codes can store different types of data, such as URLs, product details, contact information, or even payment instructions. They are easily scanned using smartphones or QR code readers, making them a popular method for connecting users to digital content or providing authentication in various applications.

1.4.2. Fake Product Identification System:

The **Fake Product Identification System** uses cutting-edge technologies such as Blockchain and digital signatures to prevent the proliferation of counterfeit goods. By embedding QR codes or barcodes into products, each item is assigned a unique digital signature. The product's information, including its origin, batch, and manufacturing details, is then stored in a decentralized and secure Blockchain ledger. Each time a product is scanned, the system checks the digital signature against the Blockchain to verify its authenticity. If the signature matches the stored data, the product is authenticated as genuine. However, if the signature does not align or has been tampered with, the system flags the product as potentially counterfeit. This method ensures transparency and traceability in the supply chain, making it nearly impossible for counterfeiters to manipulate product data without detection.

To enhance the reliability of the identification system, Blockchain technology's tamper-proof nature is leveraged. Blockchain stores the product data across multiple servers, making it resistant to fraud and unauthorized changes. Even if counterfeiters manage to clone barcodes or QR codes, their digital signatures will not align with the stored data, allowing consumers and businesses to identify and reject fake products instantly. By integrating such a system, businesses can significantly reduce losses due to counterfeit goods, especially in sectors like pharmaceuticals, electronics, and luxury goods.

1.4.3 Key Features of the System:

1. **Admin Module**:
    - Add product details and upload barcode images.
    - Generate digital signatures for barcodes and store them on the Blockchain.
    - View product and user details.
2. **User Module**:
    - Register and log in to the application.
    - Retrieve product details by entering the product ID.
    - Authenticate barcodes by uploading images and verifying digital signatures against the Blockchain.
3. **Blockchain Implementation**:
    - Uses Ethereum Blockchain for decentralized data storage.
    - Ensures data immutability and integrity through hash code verification.
    - Provides tamper-proof storage for product details and digital signatures.

1.4.4 Benefits of Blockchain Technology:

- **Tamper-Proof Storage**: Data stored in the Blockchain cannot be altered without detection, ensuring its reliability.
- **Decentralized Verification**: Eliminates the need for intermediaries, reducing costs and increasing trust.
- **Improved Security**: Cryptographic algorithms protect data from unauthorized access and manipulation.
- **Transparency**: Provides a transparent system where all stakeholders can verify product authenticity.
- **Real-Time Monitoring**: Enables instant detection of counterfeit products or fraudulent transactions.

Applications:

1. **Supply Chain**: Ensures product authenticity at every stage of distribution.

2. **Healthcare**: Prevents the entry of counterfeit medicines into the market.
3. **E-Commerce**: Provides secure and reliable transaction mechanisms.

By integrating Blockchain technology into product authentication systems, this project aims to address the critical issue of counterfeit products and fraudulent transactions. The system's decentralized and secure nature ensures that product data remains tamper-proof and trustworthy, fostering confidence among users and stakeholders. With applications across various industries, this project demonstrates the transformative potential of Blockchain in enhancing supply chain transparency and security.

1.5 Existing System:

The existing systems for product authentication and transaction verification largely rely on centralized databases and third-party intermediaries. These systems use barcodes and product labels to verify authenticity, but the data is stored in centralized servers that are vulnerable to tampering and cyberattacks. Additionally, intermediaries play a crucial role in transaction validation, which introduces potential risks of fraud and misuse of data.

Disadvantages of Existing System:

1. **Centralized Vulnerability**: Centralized databases are prone to hacking, tampering, and data breaches.
2. **Counterfeit Risk**: Barcodes and product labels can be easily duplicated, allowing fake products to enter the supply chain.
3. **Dependency on Intermediaries**: Relying on third-party intermediaries for transaction verification increases the risk of fraud and misuse of user data.
4. **Lack of Transparency**: Users and stakeholders have limited visibility into the data verification process.
5. **Slow Verification**: The involvement of intermediaries can delay the authentication and verification processes.

CHAPTER 2

LITERATURE SURVEY

Issues such as service duplication, inadequate departmental coordination, and a lack of standardization often plague supply chain management due to a lack of transparency. Counterfeiting has become increasingly prevalent, making it extremely difficult to visually identify counterfeit products. Counterfeiters cause significant challenges for legitimate firms, yet far too many people have no idea of the entire amount of counterfeit items' influence on brands. Various approaches have been developed in the past to address the challenge of product counterfeiting. The most popular methods are using RFID tags, Artificial Intelligence, QR Code based systems, etc. Each of them had a few disadvantages such as the QR Code can be copied from a genuine product and placed on a fake product, artificial intelligence uses CNN and machine learning which needs heavy computational power and so on. The idea of this paper is to improve detection of fake products by linking product SN with customer ID in the seller's system. This is achieved with Blockchain technology which ensures the identification and traceability of real products through QR Code. Blockchain based systems, makes everything decentralized. One of its main advantages is that the recorded data is difficult to change without the consent of all parties concerned which makes the data extremely secure and protected from all vulnerabilities [1].

Product counterfeiting and theft are on-going problems in supply chains and retail environments, but not a lot of work has been done to address these problems through the cost-effective use of auto-identification technologies such as bar-codes, near-field communication (NFC), or radio-frequency identification (RFID). In this paper, we propose an RFID-based anti-counterfeiting and anti-theft scheme that can be used to detect counterfeit items at the point of purchase by a consumer. The proposed system is lightweight and suited for deployment in large-scale retail environments using low-cost passive tags. We also undertake an analysis of a recent scheme proposed by Tran and Hong to highlight some of the weaknesses of their scheme. A detailed security analysis of the proposed scheme shows that it satisfies the formal requirements of security correctness and is resistant to compromise through security attacks[2].

In the paper author implemented a novel RFID-based anticounterfeiting and anti-theft scheme that is suitable for largescale implementation in retail environments. The proposed scheme is lightweight and suitable for implementation using low-cost passive RFID tags. We analysed Tran and Hong's anti-counterfeiting protocol and addressed some of the weaknesses of their scheme in Section III before proposing our novel approach. Later, we showed through a detailed security analysis that the proposed scheme was correct, satisfying the authentication freshness guarantees, and was resistant to security attacks such as database spoiling and DoS attacks. In the future, we intend to extend our work to accommodate more retail use cases such as reselling and product return scenarios also we intend to conduct another security verification using AVISPA tool [48], which is a push-button tool for the automated validation of security protocols and applications to test and add extra experimental horizon to our scheme[2].

Blockchain has gotten a lot of publicity recently, and there are a lot of applications that have come out of it. The cryptocurrency, Bitcoin is a well-known Blockchain technology which effectively solves the double-spending issue, can validate the validity of transactional history without depending on a centralised framework to do the same. As a result, the application which uses Blockchain technology as its foundation guarantees that the data is tamper-free. The paper employs a decentralised Blockchain based proposal to verify that customers do not solely depend on retailers to decide whether or not goods are original. We define a decentralised Blockchain framework without fake goods, so that producers can use it to produce legitimate products without directly handling stores, lowering product quality assurance costs significantly[3].

Counterfeit items have become more prevalent in the industrial industry in recent years. This has an effect on the company's name, sales, and profit. Blockchain technology is used to authenticate and identify counterfeit items. Using blockchain technology, transactions are recorded in the form of blocks in a distributed, decentralized, and digital ledger. A block cannot be changed or hacked since blockchain technology is secure. Blockchain technology eliminates the need for customers or users to rely on third parties to verify the product's safety. In light of current advancements in mobile and wireless technologies, quick response (QR) codes provide a promising technique for addressing the practice of counterfeiting in

this project. This is done by scanning a QR code, which is linked to a Blockchain, to identify counterfeit goods. As a consequence, this method may be used to record product information and generate a unique identifier for each product in a database. It looks up the user's unique code in the Blockchain database and sees if there are any matches. When a code matches, the client will be notified; otherwise, the consumer will be notified that the product is counterfeit[4].

In recent years, counterfeit products have played an important role in the product making industry. It has also affected the company's brand, sales, and profits. Blockchain technology is now being used to identify authentic products and detect counterfeits. Blockchain is a distributed, expanded, digital book that stores business data as blocks in multiple archives linked by chains. No block can be changed or modified because blockchain technology is secure. Customers or users who use the blockchain technology does not have to be depend on mediator to ensure product security of the system. Counterfeit products can be effectively combated using serial numbers which is assigned to the product uniquely. With emerging Blockchain technology trends in this project. As a result, this system can be used to store product information as well as generate distinct product codes for items such as blocks. It generates a unique code for the user and matches it to the Blockchain network. If the input matches, the customer will be notified; if the input does not match, the customer will be notified that the product is counterfeit[5].

Lately, fake items have taken on an undeniably significant role in the businesses that make different items. This affects the standing of the organization, as well as their deals and benefits. The execution of blockchain innovation takes into consideration the recognizable proof of certified items as well as the recognition of fake products. The blockchain is a sort of dispersed, decentralized, and computerized record that keeps value-based data as blocks in various data sets that are associated with the chains. Blockchain innovation is otherwise called Distributed Ledger Technology (DLT). Because of the way that blockchain innovation is a protected innovation, any block can neither be modified nor be hacked. Clients never again need to depend on the affirmation of item security given by outsider because of the use of Blockchain innovation. Quick Response (QR) codes are an incredible asset that can be used in this undertaking to battle the broad issue of item fraud. These codes exploit ongoing advancements in portable and remote advancements. The utilization of a Quick

Response code scanner, in which the Quick Response code of the product is connected to a Blockchain, is the technique by which counterfeit merchandise are recognized and dispensed with. Therefore, this method has the potential to be utilised in order to keep product particulars and the created unique code of that product in the database in the form of blocks. It does a comparison of the code with the entries in the Blockchain database after it has obtained the user's unique code and stored it. The consumer will receive a notification indicating the product is authentic if the code is a match; otherwise, the customer will receive a notification that the product is counterfeit[6].

The rise of counterfeit products poses a significant challenge for manufacturers, affecting their brand reputation, sales, and profitability. However, blockchain technology provides a solution by ensuring the identification and traceability of genuine products throughout the supply chain. Blockchain is a decentralized digital ledger that securely stores transactional data in interconnected databases. Leveraging blockchain for product authentication and counterfeit detection offers a tamper-proof system. One effective approach is to utilize Quick Response (QR) codes linked to the blockchain. QR code scanners verify product authenticity by comparing the scanned code with entries in the blockchain database. If a match is found, customers receive confirmation of the product's genuineness. If no match is detected, customers are promptly notified of a counterfeit product, allowing manufacturers to address the issue. Counterfeit products not only harm manufacturers but also jeopardize consumer safety, revenue, brand reputation, and trust. Establishing a reliable system to detect counterfeit products is crucial. Blockchain-based systems provide a decentralized and secure platform, ensuring product authenticity and mitigating counterfeiting risks. By leveraging blockchain and QR codes, manufacturers can protect their brands, boost consumer confidence, and maintain a secure supply chain[7].

Counterfeit items are a serious hazard to consumers and businesses, resulting in financial losses, brand reputation harm, and severe health and safety issues. In answer to this issue, we provide a revolutionary counterfeit detection system based on blockchain technology. Our project combines Kotlin, Android, Hyperledger Fabric, and FirebaseDB to create an Android app that enables users to authenticate items via barcodes, QR codes, and serial numbers. Users receive immediate feedback on the legitimacy of the goods after scanning or inputting the product details into our app.

This data is collected from a blockchain-based logging system that securely records and monitors each product's status changes. We secure the immutability and transparency of the logged data by employing blockchain technology, prohibiting unauthorized duplicates or changes of product information. Our project's implementation addresses the shortcomings of existing anti-counterfeiting techniques by offering a user-friendly and efficient means for customers to check the authenticity of their acquired items. Furthermore, the incorporation of Hyper-ledger Fabric, a permissioned blockchain platform, improves the system's security and privacy. Our research emphasizes the benefits of using Kotlin and Android development for mobile-based verification systems, as well as the potential of blockchain technology in combatting counterfeit items. Our project's results demonstrate the efficacy and feasibility of the suggested solution, presenting a potential method for protecting consumers and companies from the hazards connected with counterfeit items[8].

Counterfeit products have become a significant problem for small and medium-sized businesses (SMBs), with the estimated value of counterfeit goods worldwide reaching trillions of dollars. However, SMBs often lack the resources and technical expertise to implement sophisticated anti-counterfeiting measures. Towards this end, the work proposes a blockchain-based solution named as Fake Product Identification for Small and Medium Firms (FPISMF) using Hyperledger and AES encryption to enable SMBs to identify fake products and protect their brand reputation. The details of the products and the details of customers are encrypted using AES encryption and recorded on the blockchain. The application communicates with the blockchain network to validate the product and retrieve the details of the product. Chaincode is executed in a containerized environment, which provides isolation and security for the code and data being processed. An algorithm is also proposed to substitute the missing QR-code bits and data that helps reduce customer wait time. Experiments are conducted on synthesized data sets and results showing the effectiveness of the proposed FPISMF framework and reconciliation technique. It is observed from the results that though the time taken to replace a blurry bit is greatly reduced as compared to manual replacement of the product, there is an increase in this time when associated with encryption while extraction of the corresponding code from cloud database thereby achieving a time complexity of O(n), where 'n' is the number of scanned products. In addition, the AES SMB time complexity is approximately

recoded as O(n/2) and the Cloud access and retrieval time is O(n) as compared to O(2n) in the existing work. This shows a significant improvement in the ability to replace missing bits and perform a secure analysis respectively[9].

The field of supply chain management has long grappled with problems regarding redundancy in services, lack of coordination among various industries, and a lack of standardization, mainly due to the inherent lack of transparency. Nowadays, the proliferation of synthetic products is a rampant issue, making it incredibly challenging to distinguish a genuine item from a counterfeit one through visual inspection alone. Counterfeit goods present substantial challenges for legitimate businesses, yet the full extent of their impact on brands often remains underestimated by the public. Over time, many methods have been devised to combat the problem of product counterfeiting, including popular techniques like tags, A. I., and bar code-based systems. However, each of these approaches has its limitations, such as the vulnerability of QR codes to copying and placement on fake products, or the resource-intensive nature of artificial intelligence, which relies on complex computational power. The central idea is to enhance the detection of counterfeit products by effectively tracing its entire logistics network. This can be achieved through the application of blockchain technology, which ensures the tracking of authentic products throughout the entire network. A Blockchain systems decentralize data access, allowing multiple parties to access it simultaneously. A key advantage of this technology is that the registered data is exceptionally resistant to alteration without the unanimous consent involved, making it highly secure and protected against vulnerabilities. In summary, this paper presents a system designed using blockchain technology to improve the detection of synthetic products by providing a comprehensive and secure means of tracking their journey through the logistics network[10].

In recent years, Counterfeit products play a vital role in product manufacturing industries. This Phenomenon affects the sales and profit of the companies. To ensure the identification of real products throughout the supply chain, functional blockchain technology is used for preventing product counterfeiting. By using blockchain technology, consumers do not need to rely on trusted third parties to know the source of the purchased product safely. In this paper, counterfeit products are detected using a barcode reader, where a barcode of the product is linked to a Blockchain Based

Management (BCBM) system. So, the proposed system may be used to store product details and the unique code of that product as blocks in the database. It collects the unique code from the customer and compares the code against entries in the blockchain database. If the code matches, it will give a notification to the customer, otherwise it gets information from the customer about where they bought the product to detect counterfeit product manufacturers.[11]

This paper [12] presents a modern and convenient phenomenon using the Blockchain and Supply Chain technologies which itself dispenses high security and transparency in the system, but to escalate these features some extra characteristics are added in this study which is using the One Time Password (OTP) authentication for verifying the legitimate supply chain members and products, and updating the product details in the blockchain after it is sent to the next stage in the supply chain, and further the product standards are monitored by the Quality Control Officer who is deployed by the factory in-charge for the same. Taking inspiration from the related works of the researchers who have developed various creative models which have been of great use to the community in preventing the counterfeit of products in different industries. In the future, it is believed that this methodology can be made a base and many more features will be added to it including a mobile application that can be installed on the mobile phones of the supply chain members for a smoother experience for OTP verification and tracing-and-tracking the product easily making the system even more transparent and stable.

Reading product reviews before buying the product becomes a habit[13], especially for potential customers. For a company, the positive reviews from customers can generate significant financial benefits for the business, which can be taken as input for decisions related to product design and what services are provided to customers. Related to the financial benefits gained as a result of the positive reviews about the product or service from the customer, the fraudsters tried to play the existing system by writing fake reviews and providing an assessment that is not fair to promote or discredit a product or service. Automatic detection of spammers is a very important task but still lacks research. Unlike other types of spam, such as web spam or email spam, spam on a review is far more difficult to detect. The main reason is that spammers can easily disguise themselves. Thus, it is difficult for users to recognize. After the financial benefits gained as a result of the positive reviews about

the product or service from the customer, the fraudsters tried to play with the existing system by writing fake reviews and providing an assessment that is not fair to promote or discredit a product or service for a company, the positive reviews from customers can generate significant financial benefits for businesses, it can be treated as an input for decisions related to product design and what services are provided to customers

Counterfeit items have become more prevalent in the industrial industry in recent years. This has an effect on the company's name, sales, and profit. Blockchain technology is used to authenticate and identify counterfeit items. Using blockchain technology, transactions are recorded in the form of blocks in a distributed, decentralized, and digital ledger. A block cannot be changed or hacked since blockchain technology is secure. Blockchain technology eliminates the need for customers or users to rely on third parties to verify the product's safety. In light of current advancements in mobile and wireless technologies, quick response (QR) codes provide a promising technique for addressing the practice of counterfeiting in this project. This is done by scanning a QR code, which is linked to a Blockchain, to identify counterfeit goods. As a consequence, this method may be used to record product information and generate a unique identifier for each product in a database. It looks up the user's unique code in the Blockchain database and sees if there are any matches. When a code matches, the client will be notified; otherwise, the consumer will be notified that the product is counterfeit. [14]

The availability of fake product in the Market is one of the biggest challenges of the online retail industry. These products appear to be genuine but they are imitations of the original branded products. Almost 20% of the products sold on online websites are fake. In recent times, block chain is receiving more engagement and various applications are been emerged from this technology. In this paper, to ensure that consumers need not depend on the distributers to know whether their products are authentic or not, we are using the decentralized Block chain technology approach. We describe a decentralization Block chain network with anti-counterfeiting items, which allows producers to deliver items without having to run clear outlets, lowering product quality assurance costs dramatically.[15]

CHAPTER 3

PROPOSED METHOD

3.1 Proposed Methodology:

In this project we are using Blockchain technology to authenticate products as these products may be supplied from multiple third-party distributors and these distributors can make clone/fake/counterfeits of this product BAR CODE and then manufacture fake products and add this counterfeit label to fake product and these fake products can cause huge loss of financial and lives if fake medicine manufacture.

Not only supply chain any other online transaction require third party to complete transaction and peoples has to trust on third parties to complete their transaction and sometime this third parties can make fraud transaction or misuse user data.

To avoid these problems, we are using Blockchain technology which does not require any third party and verification will be done by software algorithm itself without involvement of any third party. In this to avoid fake counterfeit we are converting all products details/barcode into digital signatures and these digital signatures will be stored in Blockchain server as this Blockchain server support tamper proof data storage and nobody can hack or alter its data and if by any chance if its data alter then verification get failed at next block storage and user may get intimation about data alter.

In Blockchain technology same transaction data stored at multiple servers with hash code verification and if data alter at one server, then it will detected from other server as for same data hash code will get different. For example, in Blockchain technology data will be stored at multiple servers and if malicious users alter data at one servers then its hash code will get changed in one server and other servers left unchanged and this changed hash code will be detected at verification time and future malicious user changes can be prevented.

In Blockchain each data will be stored by verifying old hash codes and if old hash codes remain unchanged then data will be considerd as original and unchanged and then new transaction data will be appended to Blockchain as new block. For each new data storage all blocks hash code will be verified.

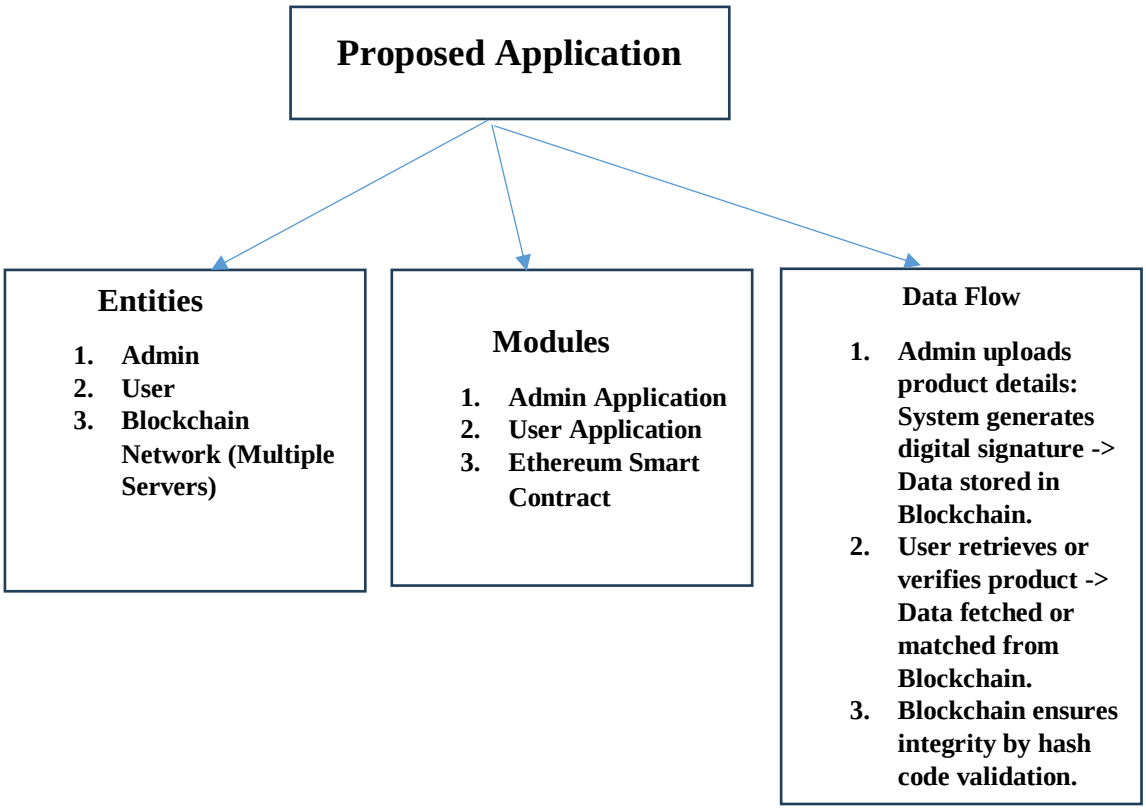


Fig.3.1 Flowchart of Proposed method

The project follows an agile methodology with iterative development and testing. Key steps include:

1. **Requirement Analysis:**
    - Understanding the problem of counterfeit products and designing a Blockchain-based solution.
2. **System Design:**
    - Designing the architecture, data flow, and module interactions.
3. **Development:**
    - Implementing modules using Ethereum Blockchain, Solidity smart contracts, and a user interface.
4. **Testing:**
    - Verifying each module's functionality and ensuring the accuracy of Blockchain-based authentication.

5. **Deployment:**
    - Deploying the system and training users for effective utilization.
6. **Maintenance:**
    - Continuously monitoring and updating the system for performance improvements.

3.2 Data Collection:

1. **Product Information:**
    - Details such as product name, ID, manufacturer, and barcode.
2. **Barcode Data:**
    - Images of barcodes for generating digital signatures.
3. **User Data:**
    - User registration details for authentication and monitoring.
4. **Blockchain Data:**
    - Digital signatures and hash codes stored securely in the Blockchain.

Data is collected from trusted sources, including product manufacturers and distributors, to ensure authenticity and accuracy.

3.3 Advantages of Proposed System:

1. **Tamper-Proof Data**: Blockchain's decentralized and immutable nature ensures that data cannot be altered without detection.
2. **Elimination of Intermediaries**: Transactions are verified by the Blockchain's algorithms, reducing dependency on third parties and minimizing fraud risks.
3. **Enhanced Security**: Cryptographic techniques protect data from unauthorized access and manipulation.
4. **Transparency**: All stakeholders can access and verify product data in real time, fostering trust.
5. **Scalability**: The system can be implemented across various industries, including supply chain, healthcare, and e-commerce.
6. **Efficiency**: Automated verification reduces delays and ensures real-time authentication of products and transactions

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 Functional Requirements:

1. **Admin Features:**
    - Admin login and authentication.
    - Add, view, and manage product details.
    - Generate digital signatures for barcodes.
    - View registered user details.
2. **User Features:**
    - User registration and login.
    - Retrieve product details by entering product ID.
    - Scan and authenticate barcodes against Blockchain data.
3. **Blockchain Features:**
    - Store product data and digital signatures securely.
    - Ensure data immutability and decentralization.
    - Verify hash codes across multiple nodes.
4. **Verification Module:**
    - Authenticate scanned barcodes.
    - Detect and alert users of counterfeit products.

4.2 Non-Functional Requirements:

1. **Performance:**
    - The system should authenticate barcodes in real-time.
    - The Blockchain network should handle concurrent requests efficiently.
2. **Scalability:**
    - The system should support the addition of new products and users without performance degradation.
3. **Security:**
    - Data stored in Blockchain must be tamper-proof.

- User and admin credentials must be securely managed.

4. **Usability:**
    - Provide a user-friendly interface for both admin and users.
5. **Reliability:**
    - The system should ensure 99.9% uptime and data accuracy.

4.3 Software Requirement Specifications:

1. **Languages** : Python, Solidity, JavaScript
2. **IDE** : Python IDLE
3. **Editor tool** : Notepad++
4. **Ethereum tool** : Truffle
5. **Operating System** : Windows 7 or above

Ethereum Blockchain:

Ethereum is a decentralized, open-source Blockchain platform that enables the creation and execution of smart contracts and decentralized applications (dApps). It was proposed by Vitalik Buterin in late 2013 and launched in 2015. Unlike Bitcoin, which primarily serves as a digital currency, Ethereum is designed to facilitate smart contracts—self-executing contracts where the terms of the agreement are written directly into code. Ethereum uses its native cryptocurrency, Ether (ETH), to pay for transaction fees and computational services on the network. One of Ethereum's key features is the Ethereum Virtual Machine (EVM), which executes the code of smart contracts. Ethereum's Blockchain is public and decentralized, making it secure and transparent for applications ranging from finance to gaming, supply chain management, and digital identity verification.

Ethereum's ability to create custom smart contracts and execute them in a decentralized environment makes it an ideal platform for a fake product identification system. By using Ethereum’s capabilities, businesses can create immutable records of product authenticity and securely verify transactions without relying on intermediaries. Smart contracts can be designed to automatically trigger actions, such as notifying consumers or businesses when a counterfeit product is detected.

Blockchain Ethereum Tool for Solidity Programming:

Solidity is the primary programming language for developing smart contracts on the Ethereum platform. It is a statically-typed, contract-oriented language that is designed to be simple to use and secure for creating decentralized applications. Solidity is a high-level language similar to JavaScript and C++, and it allows developers to write code that can execute logic on the Ethereum Blockchain. Solidity smart contracts are deployed to the Ethereum Virtual Machine (EVM), where they are executed in a decentralized and transparent manner.

When developing a fake product identification system using Blockchain, **Solidity** can be used to write smart contracts that will authenticate and verify product information stored on the Blockchain. These smart contracts can automate various aspects of product verification, such as checking if a product's digital signature matches the data stored on the Blockchain or triggering alerts when counterfeit products are detected. Additionally, tools like **Truffle** (a development framework for Ethereum) and **Remix** (an online IDE for Solidity) can assist in writing, testing, and deploying smart contracts. Truffle provides a suite of tools for Ethereum development, including automated testing and contract deployment, while Remix is an easy-to-use tool for experimenting with Solidity code directly in the browser.

By using Ethereum and Solidity, developers can create a secure and transparent system that allows businesses and consumers to easily identify fake products through Blockchain-based verification, ensuring the authenticity of products and protecting against fraud.

4.4 Hardware Requirement Specifications:

1. **Hard Disk** : 20GB
2. **RAM** : 4GB and above
3. **Processor** : i3 and above

CHAPTER 5

SYSTEM DESIGN

5.1 Block Diagram:

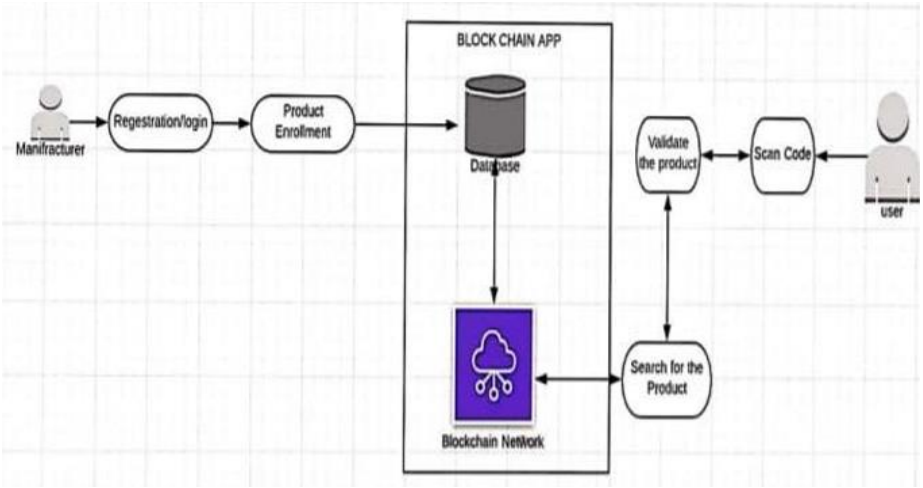


Fig.5.1 Block Diagram

This system diagram represents a blockchain-based product authentication process. The manufacturer begins by registering or logging into the system and enrolling product details into a blockchain application. These details are first stored in a database and then recorded in a blockchain network to ensure security and transparency.

On the user's side, they scan a QR code or barcode on the product. The system searches the blockchain network for the product details and performs a validation check. If the details match the blockchain records, the product is verified as authentic. If no match is found, it may indicate a counterfeit product.

This verification system helps prevent product tampering and counterfeiting while fostering trust between manufacturers and consumers. It is widely applicable in industries such as pharmaceuticals, electronics, and luxury goods, where product authenticity is a priority. The decentralized nature of blockchain enhances data integrity and security, making verification more reliable.

5.2 System Architecture Description:

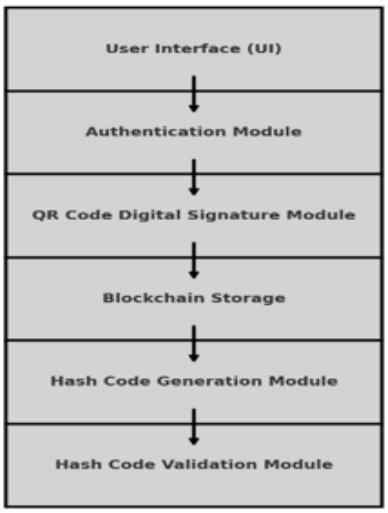


Fig.5.2 System Architecture Diagram

The system architecture for Fake Product Identification Using Blockchain is designed to ensure a secure and efficient method for product authentication. The architecture integrates various components and modules to provide a seamless and tamper-proof mechanism for verifying product authenticity and detecting counterfeit products. Below is the detailed explanation of the architecture along with the corresponding diagram.

- **User Interface (UI):**
    - Provides the interface for users to interact with the system.
    - Allows users to scan barcodes, retrieve product data, and authenticate products.
- **Admin Module:**
    - Admin can add product details, upload barcode images, and generate digital signatures.
    - Admin also manages user details and monitors product data.
- **Blockchain Layer:**
    - Ethereum-based Blockchain is used for secure data storage.

- Implements smart contracts using Solidity programming.
- Ensures data immutability and decentralization.

- **Barcode Digital Signature Generator:**
  - Converts product barcode details into unique digital signatures.
  - Ensures that any tampering with the barcode can be detected.

- **Blockchain Storage:**
  - Stores the product details, barcode digital signatures, and transaction data in blocks.
  - Each block contains a hash code of the previous block to maintain integrity.

- **Verification Module:**
  - Verifies the authenticity of scanned barcodes against the stored digital signatures. Alerts users if discrepancies are found. Alerts users if discrepancies are found.
  - Alerts users if discrepancies are found.

Workflow:

1. Admin adds product details and barcode, which are converted into digital signatures and stored in the Blockchain.
2. When a user scans a barcode, the system generates a new digital signature and compares it with the stored one.
3. If the signatures match, the product is authenticated; otherwise, it is flagged as counterfeit.
4. The system continuously validates hash codes across Blockchain nodes to maintain data integrity.

CHAPTER 6

UML DAIGRAMS

6.1 Introduction:

Unified Modeling Language (UML) diagrams are graphical representation used to visualize, specify, construct, and document, the various aspects of a software system. UML diagrams provide a standardized way to depict system architecture, behaviour, and interactions, making them an invaluable tool for software engineers, designers, and stakeholders throughout the software development lifecycle.

In this introduction, we explore the significance of UML diagrams in software engineering, highlighting their role in improving communication, facilitating analysis and design, and enhancing the overall quality of software systems.

6.2 Use Case Diagram:

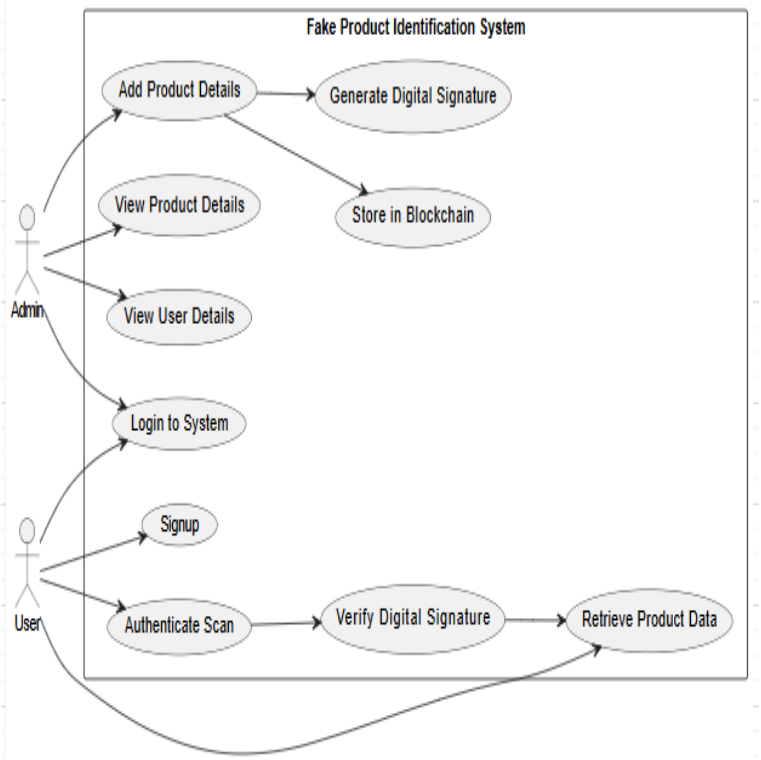


Fig.6.2 Use case diagram

The Use case diagram represents a Fake Product Identification System utilizing Blockchain technology. It showcases interactions between two key actors: Admin and User.

- The Admin can add product details, which generates a digital signature and stores the data in the blockchain. Additionally, the admin can view product details and user details.
- The User can sign up, log in, and authenticate a product scan. The system verifies the digital signature and retrieves product data from the blockchain to determine authenticity.

This system enhances transparency, security, and trust by preventing counterfeit products through blockchain-based verification.

6.3 Class Diagram:

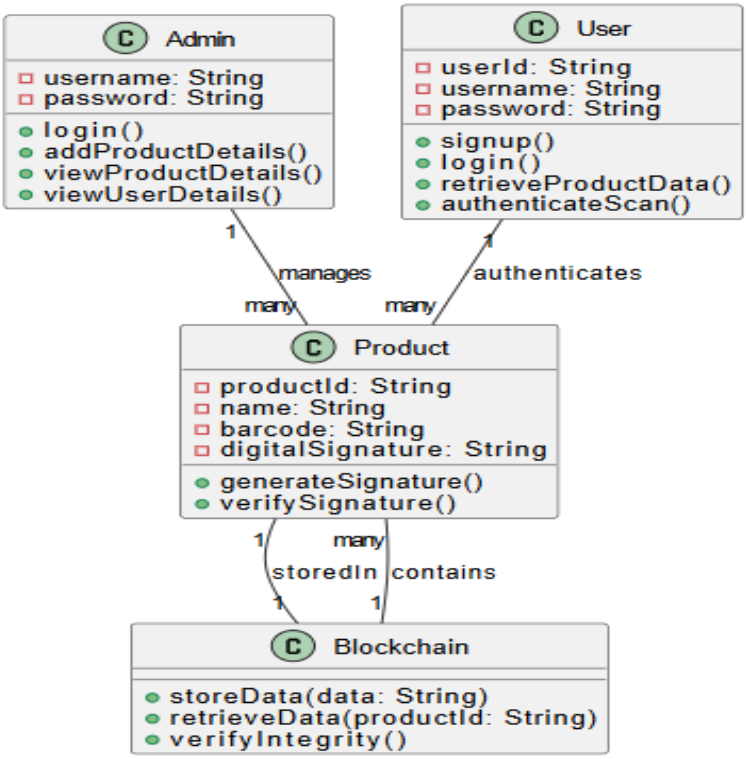


Fig.6.3 Class Diagram

The above Class Diagram represents the Fake Product Identification System using Blockchain.

- The Admin manages multiple Products, adding details and viewing user/product data.
- The User can sign up, log in, retrieve product data, and authenticate scans.
- The Product class contains attributes like product ID, name, barcode, and digital signature, with methods to generate and verify signatures.
- The Blockchain securely stores product data, allowing data retrieval and integrity verification.

The system ensures secure product authentication, preventing counterfeiting through digital signatures and blockchain verification.

6.4 Sequence Diagram:

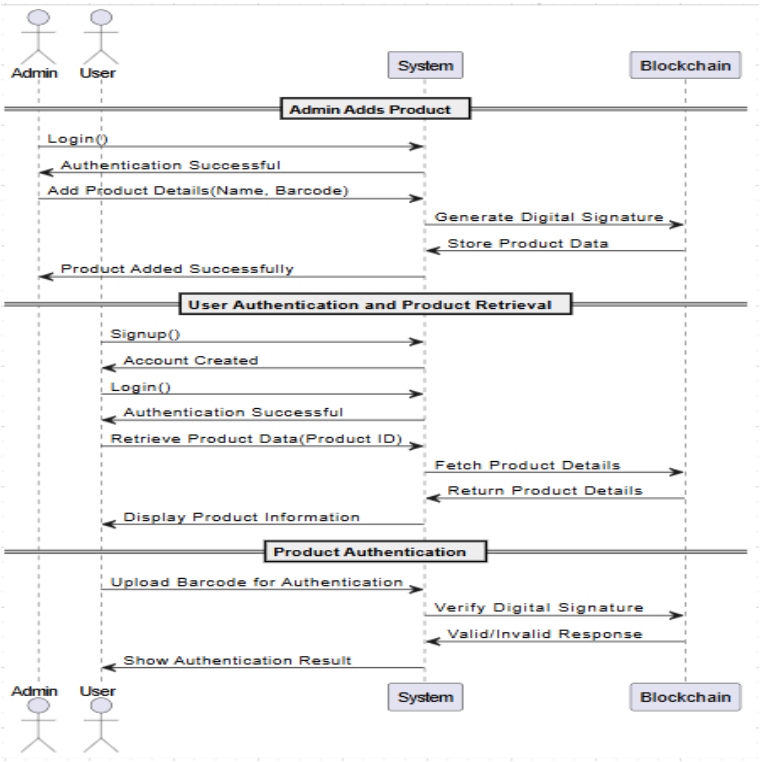


Fig.6.4 Sequence diagram

A Sequence Diagram is a type of UML (Unified Modeling Language) diagram that visualizes how objects and components interact in a specific order within a system. It is mainly used in system design and software engineering to represent message flow over time.

The above Sequence Diagram represents the Fake Product Identification System using Blockchain.

- Admin Adds Product: The admin logs in, adds product details (name, barcode), which generates a digital signature and stores data in the blockchain.
- User Authentication & Product Retrieval: The user signs up, logs in, and retrieves product data stored in the blockchain.
- Product Authentication: The user uploads a barcode, the system verifies the digital signature, and returns a valid/invalid response for authenticity verification.

This system ensures secure product verification and prevents counterfeiting using blockchain.

6.5 Collaboration Diagram:

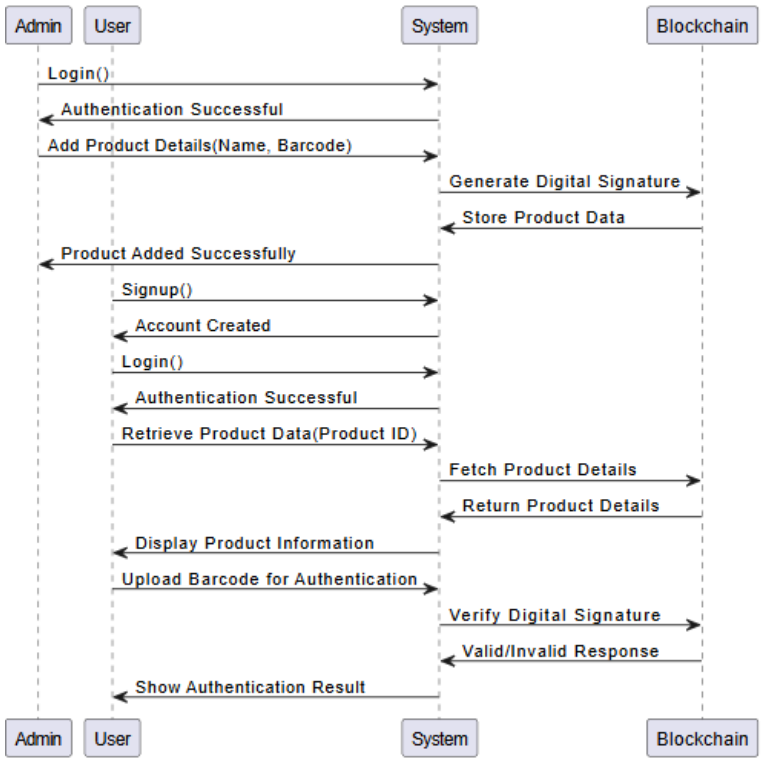


Fig.6.5 Collaboration Diagram

A Collaboration Diagram (also known as an Interaction Diagram or Communication Diagram) is a type of UML (Unified Modeling Language) diagram

that illustrates interactions between objects and components in a system. It focuses on how objects collaborate to achieve a specific functionality.

The above Collaboration Diagram illustrates the Fake Product Identification System using Blockchain.

- Admin adds a product, generating a digital signature and storing it in the blockchain.
- Users authenticate, retrieve product details, and verify authenticity via the system.
- The system fetches product data, verifies the digital signature, and provides a valid/invalid response, ensuring secure product verification.

6.6 Activity Diagram:

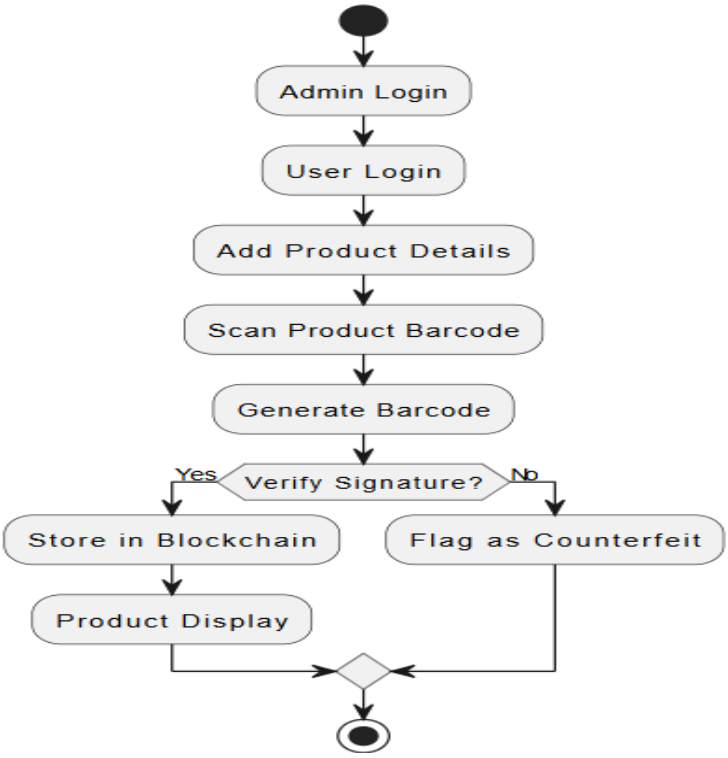


Fig.6.6 Activity Diagram

The above Activity Diagram represents the Fake Product Identification System using Blockchain for product authentication.

- Admin Login & User Login: Admin and users access the system.
- Add Product Details: Admin inputs product information.

- Scan Product Barcode & Generate Barcode: The system scans and creates a barcode

  for verification.

- Verify Signature Decision: The system checks the digital signature:

  1. If valid: The product is stored in the blockchain and displayed.

  2. If invalid: The product is flagged as counterfeit.

This Activity Diagram ensures secure product verification and counterfeit detection using Blockchain Technology.

CHAPTER 7

IMPLEMENTATION

7.1 Introduction:

Implementation is the stage of the project when the theoretical design is turned out into a working system. Thus, it can be considered to be the most critical stage in achieving a successful new system and in giving the user confidence that the new system will work and be effective. The implementation stage involves careful planning, investigation of the existing system and its constraints on implementation, designing of methods to achieve changeover and evaluation of changeover methods.

7.2 Technologies Used:

The Technologies used in creating the decentralized application:

- Python
- Cryptography
- SHA-256

Python:

Python is one of the most widely used programming languages due to its simplicity, readability, and efficiency. It is known for its versatility, making it suitable for web development, artificial intelligence, blockchain technology, and cybersecurity. With a vast collection of libraries and a strong developer community, Python provides powerful tools for handling complex tasks with minimal coding effort.

In the context of your project, Python is essential for backend development, blockchain interaction, and data security. Flask enables the creation of a web-based interface for barcode authentication and user interaction. Web3.py facilitates seamless communication with the Ethereum blockchain, ensuring secure storage and retrieval of product details. Hashlib, particularly SHA-256, is used to generate cryptographic hashes for verifying barcode authenticity.

Python's efficiency in handling APIs, managing databases, and interacting with blockchain networks makes it an excellent choice. Its scalability allows for future enhancements such as AI-driven fraud detection and automated verification processes.

Key Features of Python:

- **Web Development with Flask**: Handles API requests, user authentication, and barcode verification.
- **Blockchain Interaction with Web3.py**: Calls smart contract functions to store and retrieve product details.
- **Cryptographic Security with Hashlib**: Uses SHA-256 hashing to authenticate barcodes securely.
- **File and Database Handling**: Manages blockchain transaction records and barcode files efficiently.
- **Scalability and Flexibility**: Can be extended to include AI-based fraud detection in the future**.**

Cryptography:

Cryptography is essential for ensuring data security, privacy, and integrity in modern digital systems. It protects sensitive information by converting it into an unreadable format, making it accessible only to authorized users. Cryptography is widely used in areas such as secure communication, authentication, digital signatures, and blockchain technology.

One of the primary roles of cryptography is confidentiality, which ensures that data remains private and cannot be accessed by unauthorized parties. It also provides data integrity, preventing unauthorized modifications by detecting tampering through cryptographic hashes. Authentication is another key function, verifying the identities of users and systems through secure encryption mechanisms.

In blockchain technology, cryptography plays a crucial role in securing transactions and maintaining trust. Hashing algorithms like SHA-256 are used to secure blockchain data, while public-key cryptography enables secure digital signatures and wallet encryption. Cryptography ensures the safety of sensitive data in financial transactions, password storage, and secure messaging applications.

Features of Cryptography:

- **Confidentiality:** Ensures that only authorized users can access sensitive data by encrypting information.
- **Data Integrity:** Prevents unauthorized modifications by detecting changes through cryptographic hashing.
- **Authentication:** Verifies user identities and system access using encryption mechanisms like digital signatures.
- **Encryption and Decryption:** Converts plaintext into ciphertext to protect data and restores it when needed.

SHA-256:

SHA-256 (Secure Hash Algorithm 256-bit) is a cryptographic hash function that generates a unique 256-bit (64-character hexadecimal) hash value for any input data. It is part of the SHA-2 family and is widely used for securing sensitive information in applications like blockchain, digital signatures, and password hashing. SHA-256 is known for its strong security, as it is resistant to collisions, meaning two different inputs will not produce the same hash. It is also irreversible, making it computationally infeasible to determine the original input from the hash.

In this project, SHA-256 plays a crucial role in ensuring the security and authenticity of barcode data. When a product barcode is uploaded, its contents are read and converted into a SHA-256 hash. This hash acts as a digital signature, ensuring that the barcode data cannot be altered without detection. The hashed value is then stored on the blockchain using Web3.py and smart contracts. When a barcode is later scanned for verification, its SHA-256 hash is computed again and compared with the stored hash. If the values match, the barcode is authenticated, preventing fraud and ensuring product integrity. This cryptographic approach secures barcode verification against tampering and duplication.

Features of SHA-256:

- **Fixed-Length Output**: SHA-256 always produces a 256-bit (64-character hexadecimal) hash, regardless of input size.

- **Irreversibility**: It is computationally infeasible to reverse the hash to obtain the original input data.
- **Collision Resistance**: No two different inputs can produce the same hash, ensuring data integrity.
- **Deterministic Hashing**: The same input will always generate the same hash, making verification reliable.
- **Fast and Secure**: Optimized for efficiency while providing strong security against cryptographic attacks.

7.3 Modules:

- **Admin Module**: Admin can login to application by using username and password as ‘admin’ and ‘admin’. After login admin will perform operations.
- **Add Product Details**: Using this module admin will enter product details and then upload Barcode image and then system will generate digital signature from Barcode image and then store all those details in Blockchain.
- **View Product Details**: Using this module admin can see all product details uploaded by him/her.
- **View User Details**: Admin can see all registered user details.
- **User Signup**: Using this module user can signup with the application.
- **User Login**: Using this module user can login to application.
- **Retrieve Product Data**: Using this module user can enter product id and then product details will be retrieve from Blockchain.
- **Authenticate Scan**: Using this module user can upload Barcode image and then this image will get authenticate from Blockchain based on Digital signatures and if Barcode is not tampered then it will get authenticated and product details will be retrieve.

CHAPTER 8

TEST CASES

8.1 Introduction:

Test cases are vital components of software testing, serving as detailed descriptions of how to verify the functionality of a software application or system. They provide a structured approach to assess whether the software meets specified requirements and functions correctly under various conditions. The creation and execution of effective test cases are essential for ensuring the quality, reliability, and usability of software products.

8.2 Types of Testing:

1. **Unit Testing:** This tests individual components like Flask routes, barcode hashing, and Web3.py functions. Python's unittest or pytest can be used to check if functions return expected outputs.
2. **Integration Testing:** Ensures that different modules (Flask, Web3.py, and smart contracts) work together. Tests will confirm that Flask properly interacts with Web3.py and retrieves blockchain-stored data.
3. **Functional Testing:** Verifies that all features, like barcode verification and blockchain storage, work as intended. This ensures that API requests return the correct data.
4. **Security Testing:** Focuses on cryptographic integrity and blockchain security. SHA-256 hashing should be tested for consistency, and blockchain interactions should be checked for vulnerabilities.
5. **Performance Testing:** Measures how well the system performs under different loads. Since blockchain transactions require processing time, testing should ensure that the response times remain acceptable.

8.3 Test Cases:

Test Case-1: Admin Test

Test Case	1
Name of Test	Admin
Input	Takes the product details and barcode
Expected Output	Generation of Digital Signature
Actual Output	Generates Digital Signature
Result	Successful

Test Case-2: User Test Case1

Test Case	2
Name of Test	User-1
Input	Uploads Original Barcode
Expected Output	Gives Product Details
Actual Output	Displays Product Details
Result	Successful

Test Case-3: User Test Case2

Test Case	3
Name of Test	User-2
Input	Uploads Fake Barcode
Expected Output	Authentication Fails
Actual Output	Displays Authentication Failed
Result	Successful

Test Case-4: Product History

Test Case	4
Name of Test	Product History
Input	Using Product ID
Expected Output	Displays Product Details
Actual Output	Displays Product Details
Result	Successful

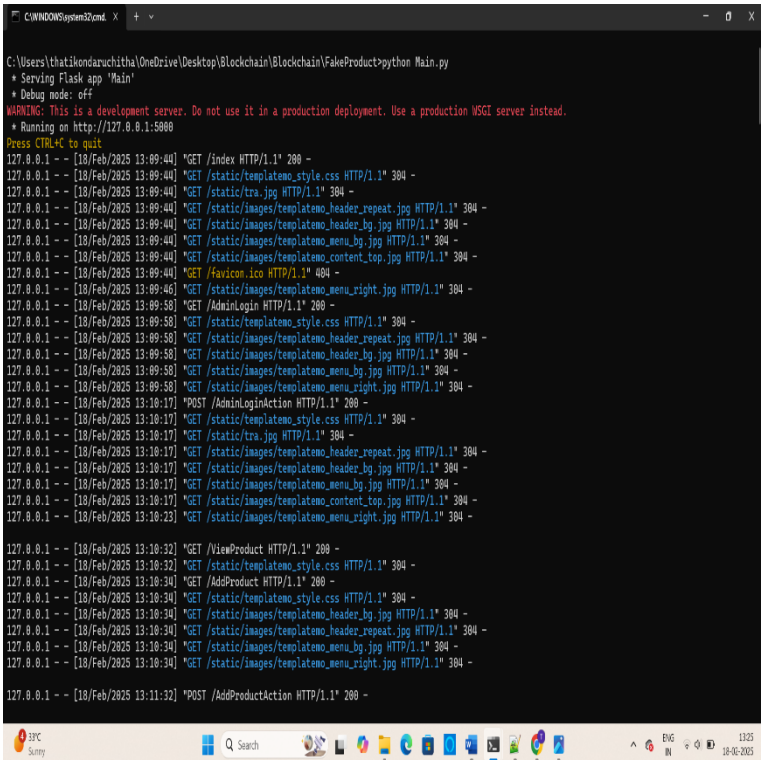
Test Case-5: Protect Against Unauthorized Access

Test Case	5
Name of Test	Unauthorized Access
Input	Username and Password Credentials
Expected Output	Admin only have the access
Actual Output	Only admin have the access
Result	Successful

CHAPTER 9

RESULTS

URL:



```
C:\Users\thatikondaruchitha\OneDrive\Desktop\Blockchain\Blockchain\FakeProduct>python Main.py
 * Serving Flask app 'Main'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /index HTTP/1.1" 200 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /static/templatemo_style.css HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /static/tra.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /static/images/templatemo_header_repeat.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /static/images/templatemo_header_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /static/images/templatemo_menu_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /static/images/templatemo_content_top.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:44] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [18/Feb/2025 13:09:46] "GET /static/images/templatemo_menu_right.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:58] "GET /AdminLogin HTTP/1.1" 200 -
127.0.0.1 - - [18/Feb/2025 13:09:58] "GET /static/templatemo_style.css HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:58] "GET /static/images/templatemo_header_repeat.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:58] "GET /static/images/templatemo_header_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:58] "GET /static/images/templatemo_menu_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:09:58] "GET /static/images/templatemo_menu_right.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "POST /AdminLoginAction HTTP/1.1" 200 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "GET /static/templatemo_style.css HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "GET /static/tra.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "GET /static/images/templatemo_header_repeat.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "GET /static/images/templatemo_header_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "GET /static/images/templatemo_menu_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:17] "GET /static/images/templatemo_content_top.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:23] "GET /static/images/templatemo_menu_right.jpg HTTP/1.1" 304 -

127.0.0.1 - - [18/Feb/2025 13:10:32] "GET /ViewProduct HTTP/1.1" 200 -
127.0.0.1 - - [18/Feb/2025 13:10:32] "GET /static/templatemo_style.css HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:34] "GET /AddProduct HTTP/1.1" 200 -
127.0.0.1 - - [18/Feb/2025 13:10:34] "GET /static/templatemo_style.css HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:34] "GET /static/images/templatemo_header_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:34] "GET /static/images/templatemo_header_repeat.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:34] "GET /static/images/templatemo_menu_bg.jpg HTTP/1.1" 304 -
127.0.0.1 - - [18/Feb/2025 13:10:34] "GET /static/images/templatemo_menu_right.jpg HTTP/1.1" 304 -

127.0.0.1 - - [18/Feb/2025 13:11:32] "POST /AddProductAction HTTP/1.1" 200 -
```

Fig.9.1 FLASK Server Started

The URL http://127.0.0.1:5000 indicates that a Flask web application is running on the local machine using the default development server. The IP address 127.0.0.1 (localhost) ensures that the server is only accessible from the same computer. The logs show multiple HTTP requests, including GET and POST methods. Requests to endpoints such as ViewProduct and AddProduct indicate product-related functionalities, while AdminLoginAction handles administrative authentication. Several requests access static files from the /static/... directory, including CSS and images.

Admin Screens:

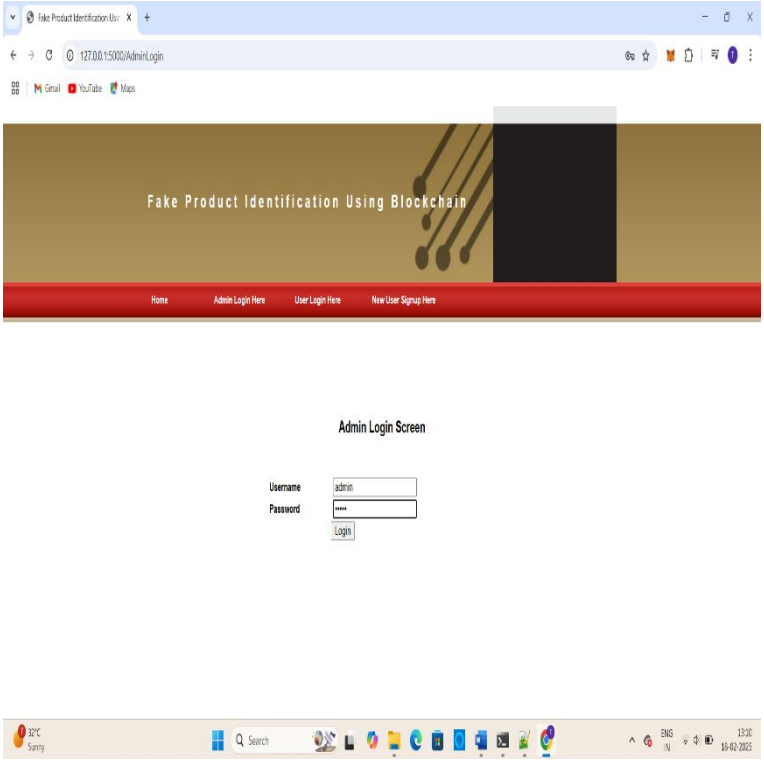


Fig.9.2 Admin Login Screen

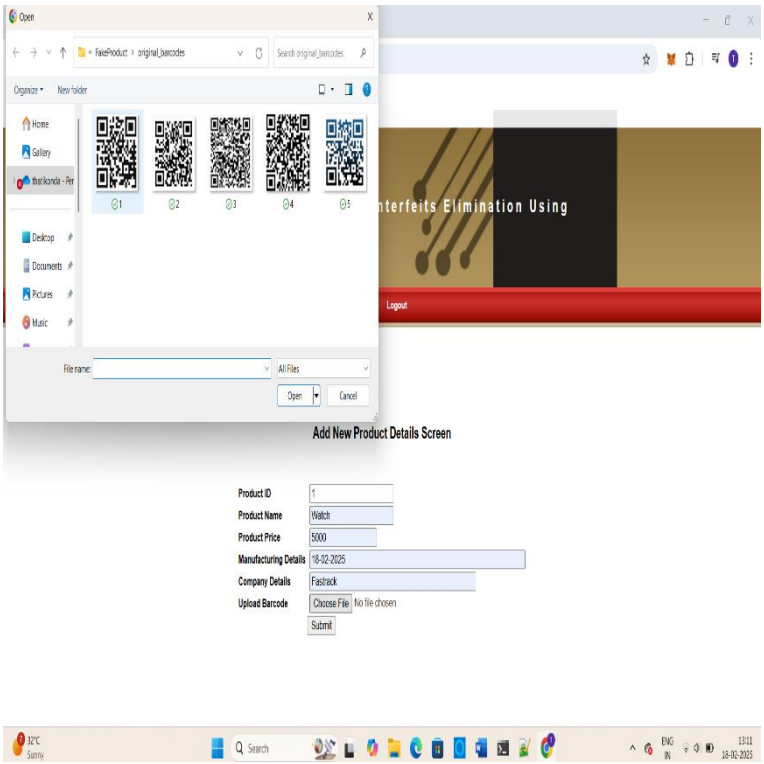


Fig.9.3 Adding Product Details

User Screens:

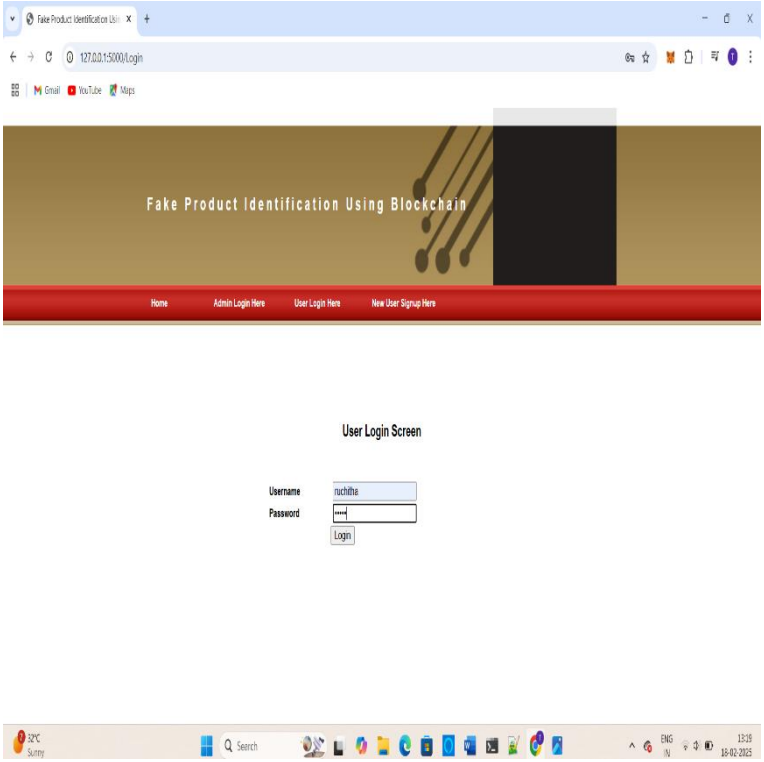


Fig.9.4 User Login Screen

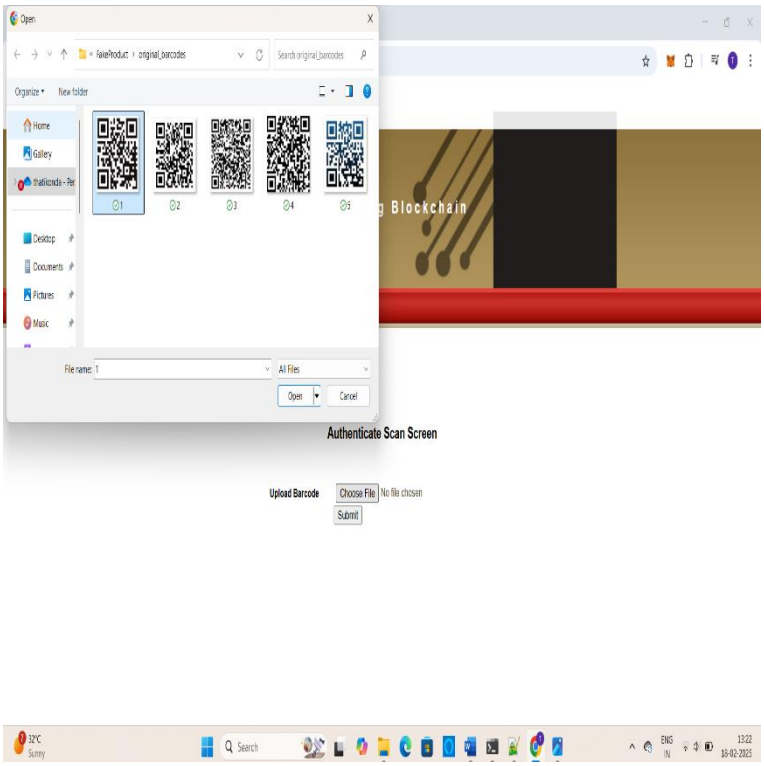


Fig.9.5 Authenticate Screen

Output Screens:

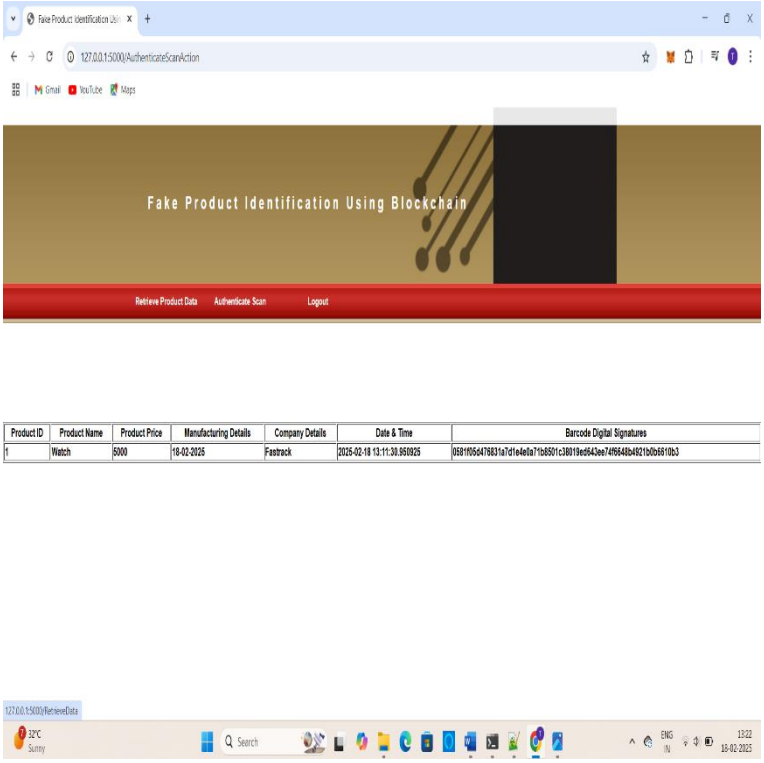


Fig.9.6 Genuine Product

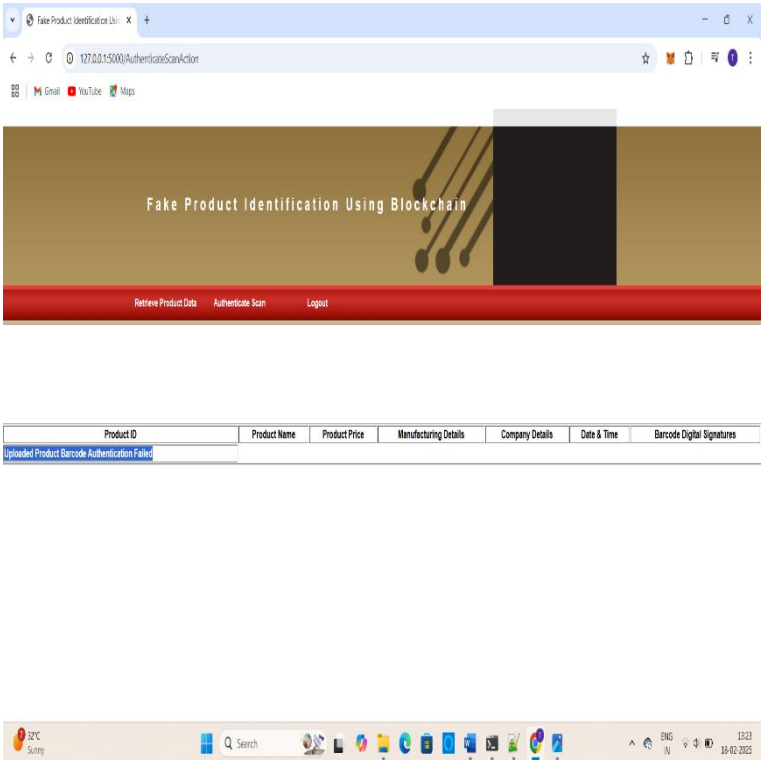


Fig.9.7 Counterfeit Product

CHAPTER 10

CODING

10.1 Introduction:

In the rapidly evolving landscape of technology and software development, code serves as the fundamental building block of virtually all digital innovations. Whether it's creating a mobile application, developing a website, or building complex systems, code is the language that translates ideas into functional software solutions.

This introduction aims to delve into the significance of code in the modern digital era, highlighting it's role in driving innovation, enabling automation, shaping the technological landscape we interact with daily.

Code forms the backbone of software development, serving as the instructions that computers follow to execute tasks and perform operations. It encompasses a wide array of programming languages, each with it's syntax, rules, and paradigms, catering to diverse application domains and development needs.

10.2 Code:

```
#Solidity Code

pragma solidity >= 0.8.11 <= 0.8.11;

contract FakeProduct {

   string public user_details;

   string public product_details;

function setUserDetails(string memory ud) public {

      user_details = ud;

   }

function getUserDetails() public view returns (string memory) {

       return user_details;

   }
```

```
function setProductDetails(string memory pd) public {
        product_details = pd;
    }
function getProductDetails() public view returns (string memory) {
         return product_details;
    }
constructor() public {
        user_details = "empty";
            product_details = "empty";
    }
}
```

This Solidity smart contract, named Fake Product, is designed to store and retrieve user and product details on the Ethereum blockchain. Let's go through it in detail.

The Fake Product Solidity smart contract stores and retrieves user and product details on the Ethereum blockchain. It defines two public string variables, user_details and product_details, with functions to set and get their values. Initially, both variables are set to "empty" in the constructor.

However, in its original form, anyone could modify the data, posing security risks. To address this, an improved version includes an owner variable (set to the contract deployer) and an only Owner modifier, ensuring only the owner can update details.

The contract utilizes Solidity 0.8.11, and a minor issue was corrected by removing public from the constructor, as it's no longer required. This contract can be useful for verifying product information, but its current implementation lacks advanced features like blockchain verification or cryptographic proofs. Enhancements like role-based access, event logging, and decentralized verification could make it more robust for real-world applications.

Python Code:

```
from flask import Flask, render_template, request, redirect, url_for, session

import json

from web3 import Web3, HTTPProvider

import hashlib

from hashlib import sha256

import os

import datetime

app = Flask(__name__)

app.secret_key = 'welcome'

global uname, details

def readDetails(contract_type):

    global details

    details = ""

    blockchain_address = 'http://127.0.0.1:9545' #Blokchain connection IP

    web3 = Web3(HTTPProvider(blockchain_address))

    web3.eth.defaultAccount = web3.eth.accounts[0]

    compiled_contract_path = 'FakeProduct.json' #FakeProduct contract code

deployed_contract_address='0xBC058E07a0807E61C6F988940A42d2D307EAdAcB

 #hash address to access counter feit contract

    with open(compiled_contract_path) as file:

        contract_json = json.load(file)  # load contract info as JSON

        contract_abi = contract_json['abi']  # fetch contract's abi - necessary to call its functions
```

```
    file.close()
 contract = web3.eth.contract(address=deployed_contract_address, abi=contract_abi)
#now calling contract to access data
    if contract_type == 'adduser':
        details = contract.functions.getUserDetails().call()
    if contract_type == 'productdata':
        details = contract.functions.getProductDetails().call()
    if len(details) > 0:
        if 'empty' in details:
            details = details[5:len(details)]
    print(details)
def saveDataBlockChain(currentData, contract_type):
    global details
    global contract
    details = ""
    blockchain_address = 'http://127.0.0.1:9545'
    web3 = Web3(HTTPProvider(blockchain_address))
    web3.eth.defaultAccount = web3.eth.accounts[0]
    compiled_contract_path = 'FakeProduct.json' #FakeProduct contract file
deployed_contract_address='0xBC058E07a0807E61C6F988940A42d2D307EAdAcB
 #contract address
    with open(compiled_contract_path) as file:
        contract_json = json.load(file)  # load contract info as JSON
        contract_abi = contract_json['abi']  # fetch contract's abi - necessary to call its
```

```
functions
file.close()
    contract= web3.eth.contract(address=deployed_contract_address, abi=contract_abi)
    readDetails(contract_type)
    if contract_type == 'adduser':
        details+=currentData
        msg = contract.functions.setUserDetails(details).transact()
        tx_receipt = web3.eth.waitForTransactionReceipt(msg)
    if contract_type == 'productdata':
        details+=currentData
        msg = contract.functions.setProductDetails(details).transact()
        tx_receipt = web3.eth.waitForTransactionReceipt(msg)
@app.route('/index', methods=['GET', 'POST'])
def index():
    return render_template('index.html', msg='')
@app.route('/Login', methods=['GET', 'POST'])
def Login():
   return render_template('Login.html', msg='')
@app.route('/AdminLogin', methods=['GET', 'POST'])
def AdminLogin():
   return render_template('AdminLogin.html', msg='')
```

CHAPTER 11

CONCLUSION AND FUTURE SCOPE

11.1 Conclusion:

The integration of Blockchain technology in product authentication provides a robust solution for combating counterfeit goods in the supply chain. By converting product barcodes into digital signatures and storing them securely on the Blockchain, the system ensures the integrity and authenticity of products, preventing fraudulent activities. The decentralized nature of Blockchain, with data verification across multiple servers, makes it highly resistant to tampering and fraud. The project successfully demonstrates how Blockchain can revolutionize product verification, especially in critical industries such as pharmaceuticals, where the consequences of counterfeit goods can be disastrous. Additionally, this technology can be extended to other areas requiring secure and transparent online transactions, offering a promising approach to fraud prevention.

11.2 Future Scope:

The future scope of blockchain for fake product identification, excluding supply chains, is promising across various sectors. In pharmaceuticals, it can ensure the authenticity of medicines, preventing counterfeit drugs that endanger lives. For luxury goods, blockchain can verify product origins, protecting brands from imitation and consumers from fraud. In electronics, it can authenticate components, reducing the risk of counterfeit parts in critical devices. Blockchain can also secure digital content like software, music, and art, preventing piracy and ensuring creators receive rightful credit. Additionally, it can be used in document verification, such as academic certificates or legal papers, to prevent forgery. By integrating AI, blockchain can enhance fraud detection, making it a versatile tool for ensuring authenticity and trust in a wide range of industries. Furthermore, blockchain can be applied to ticketing systems to prevent fake tickets, and in food safety to verify the authenticity of organic or premium products. Its tamper-proof nature ensures data integrity, making it a reliable solution for industries requiring high levels of security and transparency.

CHAPTER 12

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IJRAR.ORG E-ISSN: 2348-1269, P-ISSN: 2349-5138

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IDENTIFICATION OF COUNTERFEIT PRODUCTS USING BLOCKCHAIN TECHNOLOGY

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ABSTRACT

Blockchain technology is used to authenticate products, handling the issue of counterfeit items in the supply chain. Examples include counterfeit goods like malicious counterfeit drugs, which can cause huge amounts of monetary damage and threaten public safety. The problem occurs when third-party distributors duplicate barcodes or even create fake products, which could result in a collapse of trust for online transactions. Blockchain solves the problem by removing the requirement of third parties and it implements a tamper-proof verification system. Barcodes and other product features are transformed into digital signatures and are saved in the Blockchain to ensure their integrity. Since blockchain verifies every transaction in multiple servers, any unauthorized modification of data is detected instantly.

KEYWORDS: *Blockchain, Decentralized, Counterfeit, Ethereum*

1. INTRODUCTION

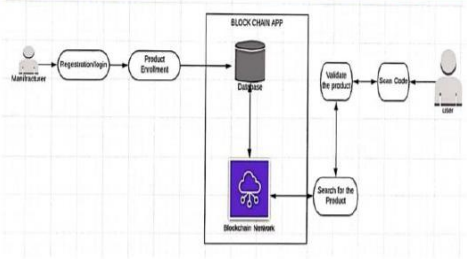
Fake products are a major issue in nearly every industry, but the supply chain, which has many distributors, is the most vulnerable. Fake products cost millions of dollars and even lives, especially in the medical industry. Legacy authentication systems easily depend on intermediaries and can be easily tampered with and abused by data. By storing product information and barcodes as digital signatures on a tamper-proof ledger, blockchain technology offers the secure, decentralized solution to this problem. No matter what you do, data will be detected immediately if altered. In this system, the Ethereum blockchain and smart contracts have been used to authenticate the products. Manufacturers help this process on their end by assigning an ID to products in the form of a QR Code, which the consumer scans to confirm authenticity. The process reduces risks of counterfeiting, safeguards, brand reputation, and increases customer trust.

2. EXISTING SYSTEM

Existing product authentication systems are built on centralized databases and third-party intermediaries. For example, barcodes and product labels verify authenticity. However, both are auxiliary and can still be hacked, tampered with, and subjected to cyberattacks, as data is stored on centralized servers. This increases the threat of fake goods making their way into the supply chain. One major limitation is reliance on intermediaries to validate transactions, which opens up the potential for fraud and misuse of information. Records are subject to the influence of malicious intermediaries who violate the security and authenticity of the records. This verification is often slow because it involves an intermediary, which delays authentication. In addition, it is very easy to duplicate barcodes and labels, which leaves us confused about which products are real and which are fake.

3. PROPOSED SYSTEM

We are utilizing blockchain technology that does not involve any third party, and verification will be performed by the software algorithm itself without any third-party involvement. To prevent counterfeit products, we are converting products details/barcodes into digital signatures and these digital signatures will be saved in Blockchain. As this blockchain supports tamper-proof data storage, nobody can hack or modify its data, and if by any chance its data gets modified then verification fail at next block storage. In Blockchain, all data will be saved by authenticating previous hash codes and if previous hash codes are not altered then data will be treated as original and then new transaction data will be added to Blockchain as new block. For every new storage of data, all blocks hash codes will be authenticated.



1. Manufacturer: The manufacturer logs into the application and registers the product using product details and its associated QR code on the blockchain to generate digital signatures.
2. User: The user can scan the product using the product QR code and verify the product, whether it is genuine or not.

SHA-256 Algorithm:

SHA-256 (Secure Hash Algorithm 256-bit) is a cryptographic hash algorithm that hashes input data and reduces it to a fixed 256-bit (32-byte) hash. SHA-256, created by the NSA, is one of the family members of the SHA-2 family of hash functions specifically designed to produce a one-way, irrecoverable, and deterministic hash, making it safe for digital signature and password hashing. SHA-256 works in a sequence of rounds of bitwise functions, modular addition, and compression functions to achieve collision and preimage resistance.

4. MODULES

1. Admin Application
2. User Application
3. Ethereum Smart Contract

MODULES DESCRIPTION:

1. Admin Application:
The Admin Application enables administrators to upload product information, create digital signatures from barcodes, and securely store them on the blockchain. It guarantees product authenticity by having tamper-proof records. Admins can view product details after adding them, and also, they can view the registered users.

2. User Application:
The User Application allows customers to check product authenticity by scanning QR codes/barcodes associated with blockchain records. Customers can access detailed product information and ensure they are purchasing genuine products. Discrepancies in digital signatures reveal counterfeiting, safeguarding consumers from counterfeit products.

3. Ethereum Smart Contract:
Ethereum Smart Contracts are computer programs that can execute automatically and securely handle transactions and product authentication on the Blockchain. They validate data integrity automatically and resist unauthorized changes. Through the removal of intermediaries, smart contracts increase security, transparency, and efficiency in product authenticity verification.

5. CONCLUSION

The use of blockchain technology for product authentication is a strong solution against counterfeit products within the supply chain. The data collected cannot be altered, thus making it a fit and proper technology for use in industries that require a high level of security. Through the encryption of product barcodes into digital signatures and the safe storage of them on the blockchain, the platform secures the genuineness and integrity of goods and ensures prevention from fraudulent operations.

6. ENHANCEMENT

Beyond supply chains, the future of blockchain technology application in identifying fake products is positive in all industries. In the pharmaceutical industry, it can help in the identification of the authenticity of drugs and avoid counterfeit drugs that may pose dangers to people's lives. In the luxury goods context, blockchain can assist in establishing product provenance to combat counterfeiting and brand and consumer protection.

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PAPER ID : IJRAR25A2655

Registration ID : 307482

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